BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SARGODHA

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022 • Paper Code: 3477 / 3472

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Q. Paper : (Objective Type)

سوالیہ پرچہ: (معروضی طرز)

QQ-1. 1022 (جماعت دہم) — سیکنڈری پارٹ II — Session 2018-20 to 2020-22 — فزکس (معروضی) — گروپ پہلا

Q. Time: 15 minutes — Total Marks: 12 — PAPER CODE 3477

Note:- You have four choices for each objective type question as A, B, C and D. The choice, which you think, is correct; fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question. Write PAPER CODE, which is printed on this question paper, on the both sides of the Answer Sheet and fill bubbles accordingly, otherwise the student will be responsible for the situation. Use of ink remover or white correcting fluid is not allowed

Q. Q-1

Q1. An object is placed at the centre of curvature of a concave mirror. The image produced by the mirror is located.

(A) (B)

(C) (D)

Q2. Combination of capacitors of ___ [illegible in source scan]

(A) (B)

(C) (D)

Q3. The combined resistance of two identical resistors connected in series is 8 Ω. Their combined resistance in parallel arrangement will be

(A) (B)

(C) (D)

Q4. The particles emitted from a hot cathode surface are

(A) (B)

(C) (D)

Q5. The S.I unit of amplitude is ___

(A) (B)

(C) (D)

Q6. ICT stands for

(A) (B)

(C) (D)

Q7. 1024 Kilobytes are equal to

(A) (B)

(C) (D)

Q8. The half life of $^{235}_{92}U$ is

(A) (B)

(C) (D)

Q9. Kinetic energy of mass spring system is maximum at

(A) (B)

(C) (D)

Q10. The example of shock absorber of the vehicles are.

(A) (B)

(C) (D)

Q11. How does sound travel from its source to your ear.

(A) (B)

(C) (D)

Q12. Total internal reflection entire light is reflected back into

(A) (B)

(C) (D)

Paper : (Essay Type)

پرچہ: (انشائیہ طرز)

QQ-2. 1022 (**جماعت دہم**) — **سیکنڈری پارٹ** II — Session 2018-20 to 2020-22 — **فزکس** (**انشائیہ**) — **گروپ پہلا** — **حصہ اول**

Q. (Marks 5×2=10) Time: 1:45 hours — Total Marks: 48 — Part I

سوال نمبر 2- درج ذیل میں سے کوئی سے پانچ اجزاء کے مختصر جوابات تحریر کریں۔ 5x2=10

i. Define Kilowatt hour, write formula to convert Electrical energy to kilowatt hour. ii. In order to measure current in a circuit why ammeter is always connected in series? iii. How many watt-hour are there in 1000 Joules? iv. State Lenz's law. v. What is difference between step-up and step-down transformer? vi. Define Fission Reaction and write its equation. vii. Write the α-decay process by equation through radium $^{226}_{88}Ra$ viii. Write difference between Atomic number and mass number.

Q. (Marks 5×2=10) Q-3

سوال نمبر 3- درج ذیل میں سے کوئی سے پانچ اجزاء کے مختصر جوابات تحریر کریں۔ 5x2=10

i. Differentiate between loudness and quality of sound. ii. What is the audible frequency range of sound for a normal human ear. iii. Write two uses of ultrasound in the field of medical. iv. Differentiate between fixed capacitor and variable capacitor. v. Write two factors which affect the ability of a capacitor to store charge. vi. Write the name of SI unit of capacitance and define it. vii. In terms of computer what is meant by hardware and software. viii. Define information and communication technology.

Q. (Marks 5×2=10) Q-4

سوال نمبر 4- درج ذیل میں سے کوئی سے پانچ اجزاء کے مختصر جوابات تحریر کریں۔ 5x2=10

i. Distinguish mechanical and electromagnetic waves. ii. Define the term transverse waves. Give example. iii. How can the refractive index find, write its unit. iv. Describe the principal focus of a convex lens by ray diagram. v. Differenciate the types of spherical mirror with diagram. vi. Write the two uses of logic gates. vii. NAND gate is the reciprocal of AND gate, Discuss. viii. Define the term thermionic emission.

Q. (Marks 9 × 2 = 18) Part II — Note: Attempt any Two Questions.

نوٹ: کوئی سے دو سوالات کے جوابات تحریر کریں۔

Q5. 5.a. What is meant by e.m.f.? Explain with the help of diagram that how we can measure potential difference in a circuit? 1+3

b. Carbon-14 has a half life of 5730 years. How long will it take for the quantity of carbon-14 in a sample to drop to one-eight of the initial quantity. 5

Q6. 6.a. Explain the electrostatic Induction with the help of an activity of plastic and aluminum rod make a lable diagram. 2+2

b. A doctor counts 72 heart beats in 1 min. Calculate the frequency and period of the heart beats.

Q7. 7.a. Explain the image formation in convex lens with the help of four figures 1×4

b. The time period of simple pendulum is 2 sec. What will be its length on the earth? What will be its length on the moon? If g_m = g_e/6 where g = 10ms^{-2}

> Note: Board typos kept verbatim: 'In terms', 'lable diagram', 'Differenciate'.

GROUP – II

Q. Paper : (Objective Type)

سوالیہ پرچہ: (معروضی طرز)

QQ-1. 1022 (**جماعت دہم**) — **سیکنڈری پارٹ** II — Session 2018-20 to 2020-22 — **فزکس** (**معروضی**) — **گروپ دوسرا**

Q. Time: 15 minutes — Total Marks: 12 — PAPER CODE 3472

Note:- You have four choices for each objective type question as A, B, C and D. The choice, which you think, is correct; fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question. Write PAPER CODE, which is printed on this question paper, on the both sides of the Answer Sheet and fill bubbles accordingly, otherwise the student will be responsible for the situation. Use of ink remover or white correcting fluid is not allowed

Q. Q-1

Q1. Example of mechanical waves is

(A) (B)

(C) (D)

Q2. The characteristics of sound by which loud and faint sounds can be distinguished is

(A) (B)

(C) (D)

Q3. For mirrors focal length is

(A) (B)

(C) (D)

Q4. One micro coulomb charge is equal to

(A) (B)

(C) (D)

Q5. The instrument which is used to measure the current is

(A) (B)

(C) (D)

Q6. D.C. motor changes electrical energy in to

(A) (B)

(C) (D)

Q7. If X=A.B then X is '1' when

(A) (B)

(C) (D)

Q8. What does the term e-mail stands for?

(A) (B)

(C) (D)

Q9. Isotopes are atoms of the same element with different

(A) (B)

(C) (D)

Q10. The relation between v, f and λ of a wave is

(A) (B)

(C) (D)

Q11. Which is the example of a Longitudinal Waves.

(A) (B)

(C) (D)

Q12. Image formed by a camera is

(A) (B)

(C) (D)

Paper : (Essay Type)

پرچہ: (انشائیہ طرز)

QQ-2. 1022 (**جماعت دہم**) — **سیکنڈری پارٹ** II — Session 2018-20 to 2020-22 — **فزکس** (**انشائیہ**) — **گروپ دوسرا** — **حصہ اول**

Q. (Marks 5×2=10) Time: 1:45 hours — Total Marks: 48 — Part I

سوال نمبر 2- درج ذیل میں سے کوئی سے پانچ اجزاء کے مختصر جوابات تحریر کریں۔ 5x2=10

i. Prove that 1kWh = 3.6 MJ ii. State Ohm's law and sketch its diagram. iii. If 0.5 C charge passes through a wire in 10s, then calculate the current. iv. What is meant by term strength of magnetic field. v. State Faraday's Law of electromagnetic induction. vi. What do you understand by the term "mutual induction" vii. What is meant by the term isotopes write the isotopes of hydrogen. viii. Define the term natural radioactivity. Write the names of radiations.

Q. (Marks 5×2=10) Q-3

سوال نمبر 3- درج ذیل میں سے کوئی سے پانچ اجزاء کے مختصر جوابات تحریر کریں۔ 5x2=10

i. How can some whales communicate hundreds kilometres away? ii. Write the audible frequency range of Bats and Mice. iii. Why the voice of women is more shrill than that of men? iv. How can you avoid explosions due to static electricity? v. Draw electric field lines for two negative point charges. vi. What is the difference between bit and byte? vii. What are browsers? Give their two examples viii. Write two services of internet.

Q. (Marks 5×2=10) Q-4

سوال نمبر 4- درج ذیل میں سے کوئی سے پانچ اجزاء کے مختصر جوابات تحریر کریں۔ 5x2=10

i. State Hooke's law and write its equation. ii. What is meant by Crest and Trough? iii. Define damped Oscillation and write its one application. iv. Differentiate between analogue and digital electronics. v. Draw a symbolic diagram for NOR gate and Write its truth table. vi. State Snell's Law and write its formula. vii. Define principle axis and focal length. viii. What is the difference between Convex and Concave mirrors.

Q. (Marks 9 × 2 = 18) Part II — Note: Attempt any Two Questions.

نوٹ: کوئی سے دو سوالات کے جوابات تحریر کریں۔

Q5. 5.a. Define an electromagnet. Make the Relay switch circuit and discuss the Electromagnet role in this case. 1+1+2

b. Half life of a radioactive element is 10 minutes. If the initial count rate is 368 counts per minute, Find the time by which, count rate reaches 23 counts per minutes. 1+1+2+1

Q6. 6.a. Indicate two dangers in use of I.C.T in Society. Write two steps to avoid these dangers. 2+2

b. Two capacitors of capacitances 6μF and 12μF are connected in parallel with a 12V battery. Find the equivalent capacitance of combination Also find the potential difference across each capacitor.

Q7. 7.a. How to find the magnification of compound microscope by ray diagram explain it? 1+3

b. A simple pendulum completes one vibration in two seconds. Calculate its length when g = 10 m s^{-2} 5